AI-Powered Smart Dustbin System

System Architecture, Real-Time Inference & Dual-Servo Actuation Mechanism (stest.py)

1. System Overview & Waste Categories

The automated waste segregation system implemented in **stest.py** performs continuous real-time classification and hands-free sorting of incoming refuse into designated bins. Targeted for deployment on an embedded **Raspberry Pi 4**, the solution combines an OpenCV camera feed, a trained lightweight Convolutional Neural Network (TensorFlow/Keras), and a dual-servo actuation mechanism driven via GPIO.

- **Biomedical Waste:** Bio-hazardous refuse, clinical swabs, contaminated dressings, and specimen packaging.
- **Medical Waste:** Clinical non-biological disposables including latex/nitrile gloves, surgical masks, and blister packs.
- **Normal (Idle):** Ambient background or empty collection tray. Strictly locks the gate; never triggers sorting.

2. Hardware Configuration & Actuator Pinout

Actuator / Sensor	Interface	Angle / State	Operational Function
Servo 1: Gate Mechanism	GPIO 17 (BCM)	Closed: **100°** Open: **0°**	Retains waste on drop platform; rotates to 0° to drop item into chute.
Servo 2: Chute Selector	GPIO 27 (BCM)	Biomedical: **50°** Medical: **150°** Normal/Idle: **100°**	Pre-aligns chute with target compartment before Servo 1 opens.
USB Web Camera	Index 0 (V4L2)	640 × 480 px @ 30 FPS	Continuous video capture of incoming waste platform.

3. Computer Vision & Inference Pipeline

Each incoming frame from the camera stream is preprocessed before inference:

1. **Acquisition:** Frame read via camera.read() at 640×480 resolution.
2. **Resizing:** Interpolated down to 128×128 pixels matching CNN input dimensions.
3. **Color Normalization:** Converted from BGR to RGB; pixel intensities scaled to float32 [0.0, 1.0].
4. **Tensor Dimension:** Expanded with batch axis to (1, 128, 128, 3).
5. **Classification:** model.predict() outputs softmax distribution; np.argmax() extracts class index and confidence percentage.

4. Finite State Machine & Sorting Execution Flow

To prevent erratic triggers from optical noise, shadows, or hand movement, **stest.py** implements a 2-state finite state machine with consecutive temporal verification:

Stage / Phase	Action Description	Duration / Criteria
1. State Detecting	Evaluates live stream. Confidence must meet threshold and classification must remain identical over consecutive frames.	Confidence ≥ **85%** Consecutive: **5 frames**
2. Chute Pre-Positioning	**Servo 2** rotates chute to target bin (50° for biomedical, 150° for medical). **Gate (Servo 1) stays closed** to prevent misdirection.	Transit: **0.7s** Settle: **1.0s**
3. Gate Release (Drop)	**Servo 1** opens to 0°. Waste drops under gravity through the pre-aligned chute into the target container.	Transit: **0.7s** Drop wait: **2.0s**
4. Gate Closure	**Servo 1** closes back to 100°, sealing the hopper platform before the chute is permitted to move.	Transit: **0.7s**
5. Chute Neutral Reset	**Servo 2** rotates back to normal default position (100°), ready for the next cycle.	Transit: **0.7s** Settle: **1.0s**
6. Wait For Normal	FSM switches to STATE_WAIT_FOR_NORMAL. Requires empty platform frames before allowing another detection, preventing duplicate cycles.	Requires: **10 consecutive normal frames**

5. Step-by-Step Decision & Actuation Pipeline

The following structured pipeline illustrates the operational decision loop executed on every camera cycle:

STEP A: Optical Sampling & Normalization	USB webcam samples 640×480 BGR image → Resized to 128×128 → Converted to RGB → Normalized by 1/255.0 to float32 tensor.
STEP B: Deep CNN Inference & Safety Thresholding	Softmax probability evaluated. If prediction is normal or confidence < 85%, Servo 1 remains locked closed (100°) and counters reset.
STEP C: Temporal Filter (5 Consecutive Detections)	If biomedical-waste or medical_waste is detected, consecutive counter increments. Action triggers only when count >= 5.
STEP D: Actuator Sequencing (Servo 2 → Servo 1 → Reset)	1. Chute Servo 2 moves to target (50° for Biomedical, 150° for Medical). 2. System waits 1.0 s for chute settling. 3. Gate Servo 1 opens to 0° for 2.0 s drop interval. 4. Gate Servo 1 closes back to 100°. 5. Chute Servo 2 returns to 100° neutral position and settles for 1.0 s.
STEP E: Cooldown & Hopper Empty Verification (10 Frames)	FSM enters STATE_WAIT_FOR_NORMAL. Requires 10 consecutive frames of clean background before re-arming the system for the next waste object.

6. Safety Controls & Fail-Safe Cleanup Protocols

A cornerstone of the **stest.py** implementation is its fault-tolerant hardware handling:

- **Normal Class Hard Lock:** The gate servo is strictly forbidden from opening when normal is detected or confidence is under 85%. The gate stays permanently closed.
- **PWM Anti-Jitter Pulse Cutoff:** Following any servo repositioning, servo.ChangeDutyCycle(0) is invoked immediately after the travel delay. This eliminates software-PWM motor chatter and overheating.
- **Guaranteed try...finally Teardown:** On user exit ('q') or exception, the system forces Servo 1 closed (100°), returns Servo 2 to neutral (100°), halts PWM timers, runs GPIO.cleanup(), and frees the camera device.

7. System Tuning Parameters Reference

Constant Identifier	Value	Engineering Rationale
`CONFIDENCE_THRESHOLD`	85%	Filters out ambiguous classifications, glare, and environmental shadows.
`REQUIRED_CONSECUTIVE_DETECTIONS`	5 frames	Temporal low-pass filter; verifies that waste is stationary and stable.
`NORMAL_FRAMES_TO_RESET`	10 frames	Ensures previous item has dropped clear before re-arming the detection loop.
`GATE_OPEN_TIME`	2.0 s	Allows sufficient clearance for items of irregular geometry or friction to drop.
`SELECTOR_SETTLE_TIME`	1.0 s	Prevents dropping waste while the chute is still in transit.
`SERVO_MOVE_DELAY`	0.7 s	Travel window for physical standard RC servo to traverse up to 180°.